ICAISC 2027: full plan and next-phase guide

Submit Thu 12 Nov. Deadline Sun 15 Nov (day after your last exam: backup only). This replaces the earlier plan files.

1. Timeline

Date	What happens	Done when
Wed 21 Oct (today)	Claude Code: the 4 "before recording" tasks (section 3). Print consent forms. Confirm people and the room. Upload the LaTeX project to Overleaf.	Laptop shows the dog's camera; raw video saves at full resolution
Thu 22 – Sat 24 Oct	**Recording** (section 2). The first 3 takes of day 1 are practice (dev). Freeze settings after them. Label each day's takes that evening, two people.	300+ labelled gestures
Thu 22 – Sat 24 Oct (evenings / teammates)	Write in Overleaf: Abstract draft, Introduction, Background, Related Work (already drafted, polish them).	Sections 1–3 read well
Sun 25 Oct	Claude Code: the "after recording" tasks: rerun everything on the videos with gaze, all tests and figures.	`results/` complete, figures in `paper/figures/`
Mon 26 Oct	Spare: one extra session if short on gestures. Check every number makes sense.	Results frozen
Tue 27 – Wed 28 Oct	Study (Economics, PS2)	
Sat 31 Oct	Write Method details, Experimental Setup, Results.	Sections 4–6 drafted
Tue 3 Nov	Write Discussion, Conclusion; finish Abstract and Intro with real numbers.	Full 12-page draft
Fri 6 – Sat 7 Nov	Teammates + a lecturer review. Fix.	Comments addressed
Thu 12 Nov	Anonymity check, references checked, PDF check. **Submit** at icaiscsystem.icaisc.eu.	Submission ID

Exam days (keep the day before free): 29 Oct, 30 Oct, 2 Nov, 5 Nov, 9 Nov, 11 Nov, 14 Nov.

2. Recording days (22–24 Oct)

Before each session

- Dog charged, laptop charged, same Wi-Fi as the dog, Unitree app closed on all phones.
- Signed consent from everyone. Give each person an ID (P01, P02, ...). Never put names in files.
- One person holds the remote and kill switch for every walking take.
- One person keeps a paper log: take name, commander ID, people present, notes.

Roles in every take: 1 commander + 2–3 bystanders. Rotate so everyone is commander at least once.

Each take is 60–90 s. Run:

```
python run_live.py --source go2 --ip <DOG_IP> --take <NAME> --scenario stationary
python run_live.py --source go2 --ip <DOG_IP> --take <NAME> --scenario walking --drive
```

Name takes like `D1_S2_r1` (day 1, scenario S2, repetition 1).

Shot list (repeat each ~4 times across the 3 days, with different commanders):

Code	Dog	What people do
S1	still	Commander faces dog and gives all 5 gestures
S2	still	Commander gestures to dog; bystanders face each other and gesture to each other the whole time
S3	still	Commander and a bystander gesture **at the same moment** (bystander to a friend)
S4	still	Commander turns to a friend, gestures at them, turns back, gestures at the dog
S5	still	Two people gesture at the dog together (one STOP, one forward)
S6	still	Bystanders walk across in front of the dog, waving at someone off-camera
W1	walking	Commander drives the dog with gestures; bystanders stand around
W2	walking	While the dog walks, a nearby bystander gestures to a friend
W3	walking	People cross between dog and commander

Rules that make the data good

- Bystanders gesture **every ~5 s, hold 2 s, hands visible to the dog**.
- Use distances from **1 m to 4 m**. Do some takes close (1–2 m) so gaze has something to see.
- Include some looking-at-the-dog-without-gesturing moments (tests the gaze-only baseline).
- Two locations if possible (lab + corridor/lobby).

Same evening: two people label every take separately:

```
python label_episodes.py takes/<NAME>                # annotator A
python label_episodes.py takes/<NAME> --annotator B   # annotator B (after task 3 below)
```

`r` = for the robot, `n` = not for the robot, `x` = bad detection.

Fallback if WebRTC won't connect: tape a phone or webcam on the dog at camera height and use `--source webcam:N`; drive with the remote. Same pipeline, same metrics.

3. Development guide (for Claude Code)

Open Claude Code in `~/go2-engagement` (it reads `CLAUDE.md` + this file). Do the tasks in order.

Before recording (today, 21 Oct)

#	Task	Done when	Prompt for Claude Code
1	**Dog connection**	Live dog camera in `run_live.py --source go2`	"Check `engage/go2_link.py` against the go2_webrtc_connect examples, fix the API calls, and test with my dog at IP ..."
2	**Full-resolution raw video**	`raw.mp4` saved at the camera's native size; processing may downscale	"Make run_live.py save raw.mp4 at native resolution and only resize the copy used for processing. Also log per-frame timestamps."
3	**Take metadata + 2nd annotator**	`--commander P03 --people P01,P03,P05 --location lab` saved in meta.json; `label_episodes.py --annotator B` writes `labels_B.json`	"Add take metadata CLI args and a second-annotator mode."
4	**Frozen settings file**	All thresholds in `params.yaml`; its hash saved in every meta.json	"Move every threshold into params.yaml and record its hash in meta.json."

After recording (Sun 25 Oct)

#	Task	Done when	Prompt
5	**Offline replay**	`replay.py` reruns perception on every `raw.mp4` and writes new logs	"Write replay.py that recomputes all features from raw.mp4 for all takes."
6	**Gaze module**	Gaze score + σ_gaze per person, σ grows as eye distance in pixels shrinks; 4th fuzzy input with rules	"Add a gaze input from MediaPipe FaceLandmarker iris landmarks on the face crop, with distance-dependent sigma, as a 4th input to the Type-1 and IT2 systems."
7	**Dev/test split**	`splits.json`: first 3 takes of day 1 = dev, rest = test	"Create splits.json and make evaluate.py report test-only numbers."
8	**Six policies**	gesture-only, nearest, **gaze-only**, **logistic regression (trained on dev)**, Type-1, IT2	"Add gaze-only and a logistic-regression baseline to methods.py and evaluate.py."
9	**Statistics**	Wilson CIs, McNemar IT2 vs T1, Cohen's κ between annotators	"Add McNemar and Cohen's kappa to evaluate.py."
10	**Analysis**	θ_E sweep curves; results split by σ (low/high), distance, scenario; noise injection; ablations (no gaze, facing only, distance only, FOU scale k)	"Add the operating curve, sigma-bin breakdown, noise test and ablations."
11	**ROS-bag distance check**	Monocular distance vs LiDAR/RealSense on the team's `social_fusion` bag; fitted σ_d(d); figure	"Run the pipeline on the bag's colour images and compare distance with LiDAR/RealSense; fit sigma vs distance."
12	**Paper figures**	Vector PDFs, 12.2 cm wide, in `paper/figures/`; LaTeX table code in `paper/tables/`	"Make paper-ready figures and LaTeX tables from results/."

Rules: never tune anything on test takes. If a setting changes after recording, it changes for every method and every take, and is rerun offline.

4. Writing (Overleaf)

1. Overleaf → New Project → **Upload Project** → `icaisc_paper.zip`. It compiles as-is.
2. Share the project with the team. Split sections:
3. Introduction, Background, Related Work: already drafted; polish them now.
4. Method: whoever knows the code best.
5. Experiments + Results: after 25 Oct, from Claude Code's tables.
6. Red `\TODO{}` = still to write. Blue `\RES{}` = numbers to fill from results. **Both must be gone before submitting.**
7. Check every `% VERIFY` reference in `references.bib` on Google Scholar.
8. Keep it anonymous: no names, no Rysera/WSO2, no website or GitHub link, no "our previous work [x]". Real author block is commented out in `main.tex` for later.
9. Aim for 12 pages including references.